SENTINEL AI SOC HOME LAB SERIES — COMMUNITY EDITION

Lab 01 — Environment Setup & VM Networking

DIFFICULTY	ESTIMATED TIME	COST
Beginner	2 – 3 Hours	$0 — all tools free
MITRE ATT&CK	**SECURITY+ DOMAINS**	**HOST OS**
TA0043 · TA0001	1.0 General Security · 3.0 Architecture	Windows 10/11 · 16GB RAM min

⚠ **Authorized use only** — All activities must be performed exclusively within your own isolated virtual lab environment. Never run these tools against any network or system you do not own or have explicit written permission to test.

LAB OVERVIEW

In this lab you will build the foundation that every other lab in this series depends on. You will create a fully isolated virtual network with two machines — an attacker and a victim — that communicate only with each other inside a private environment called **soclab**. This is the same type of controlled environment used by professional penetration testers and SOC analysts.

ℹ **Note:** Wherever this lab shows the username `rich`, replace it with your own chosen username.

LAB ARCHITECTURE

Machine	OS	IP Address	RAM	Role
Kali Linux VM	Kali Linux 2026.1	192.168.100.20	4096 MB	Attacker
Ubuntu Server VM	Ubuntu Server 22.04 LTS	192.168.100.10	3072 MB	Victim / SIEM
Sentinel AI	React + Claude API	Host machine	—	AI Triage

LEARNING OBJECTIVES

→ Install and configure **VirtualBox** as a Type 2 hypervisor on Windows

→ Import and boot a pre-built **Kali Linux VM** from a VirtualBox image

→ Install **Ubuntu Server 22.04** from an ISO with manual configuration

→ Configure an isolated **Internal Network** named soclab in VirtualBox

→ Assign **static IP addresses** to both VMs using netplan and /etc/network/interfaces

→ Verify **bidirectional connectivity** between attacker and victim using ping

→ Explain why **isolated lab networks** are essential for safe security testing

TOOLS REQUIRED — ALL FREE

Tool	Download	Notes
VirtualBox 7.2+	virtualbox.org	Select Windows hosts
Kali Linux VM Image	kali.org/get-kali	VirtualBox pre-built image (~15GB)
Ubuntu Server 22.04 LTS	ubuntu.com/download/server	Select 22.04.5 LTS — Previous releases section

LAB INSTRUCTIONS

Phase 1 — Install VirtualBox

1. Go to **virtualbox.org** and download VirtualBox for Windows hosts
2. Run the installer with default settings and complete installation
3. Launch VirtualBox Manager — you should see an empty sidebar

Phase 2 — Import Kali Linux

1. Go to **kali.org/get-kali** and download the VirtualBox pre-built image
2. Extract the downloaded archive — you will get a .vbox and .vdi file
3. In VirtualBox: Machine → Open → select the .vbox file
4. Click Settings → System → set Base Memory to **4096MB**
5. Click Settings → Network → Adapter 1 → **Internal Network** → name: `soclab`
6. Start Kali and log in: username `kali` / password `kali`

📷 Screenshot checkpoint — capture the Kali desktop after logging in

Phase 3 — Install Ubuntu Server

1. Download Ubuntu Server 22.04.5 LTS ISO from **ubuntu.com/download/server** — select Previous releases → 22.04.5
2. In VirtualBox: New → name: **Ubuntu-Victim** → attach the ISO → RAM 3072MB · CPU 2 cores · disk 40GB
3. Settings → Network → **Internal Network** → name: `soclab`
4. Boot and install: **English** → Ubuntu Server → skip network → no proxy → guided storage (entire disk) → set your username + password → Install OpenSSH: **YES** → reboot

Phase 4 — Configure Static IP on Ubuntu

1. Log in and open the netplan config file:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

2. Enter exactly — 2 spaces per indent level:

```
network:
  version: 2
  ethernets:
    enp0s3:
      addresses:
        - 192.168.100.10/24
      dhcp4: false
```

3 Save with **Ctrl+O** → **Enter** → **Ctrl+X**, then apply:

```
sudo chmod 600 /etc/netplan/00-installer-config.yaml
sudo netplan apply
ip addr show enp0s3
```

✓ You should see 192.168.100.10/24 in the output

📷 Screenshot checkpoint — capture the terminal showing 192.168.100.10

Phase 5 — Configure Static IP on Kali

1 Open the network interfaces file:

```
sudo nano /etc/network/interfaces
```

2 Add these lines at the bottom of the file:

```
auto eth0
iface eth0 inet static
  address 192.168.100.20
  netmask 255.255.255.0
```

3 Apply and verify:

```
sudo systemctl restart networking
ip addr show eth0
```

✓ You should see 192.168.100.20/24 in the output

📷 Screenshot checkpoint — capture the terminal showing 192.168.100.20

Phase 6 — Verify Connectivity

1 From Kali, ping the Ubuntu victim machine:

```
ping 192.168.100.10
```

✓ Expected: 64 bytes from 192.168.100.10 — 0% packet loss. Press Ctrl+C to stop.

📷 Screenshot checkpoint — capture the ping output showing 0% packet loss

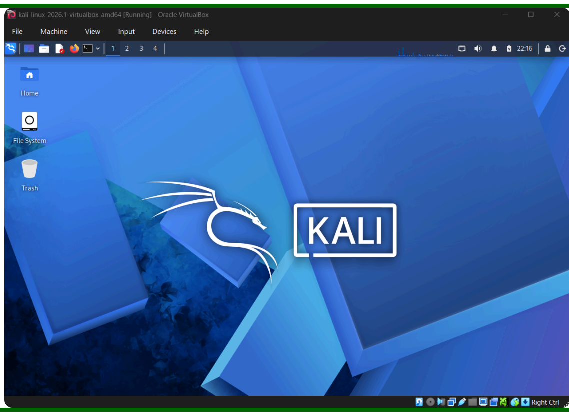
📷 Screenshot checkpoint — capture the VirtualBox sidebar with both VMs showing Running

YOUR SCREENSHOTS

Click each zone to upload your screenshot. They will appear in the document when you print or save as PDF.

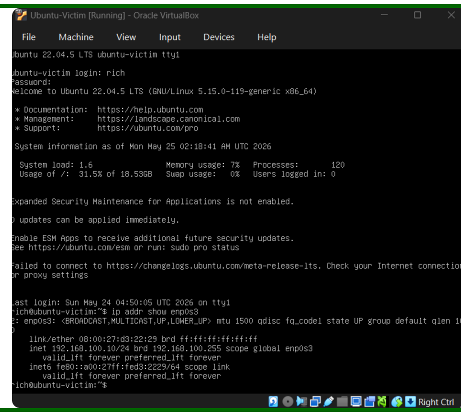
Screenshot 1

lab01_kali_desktop.png



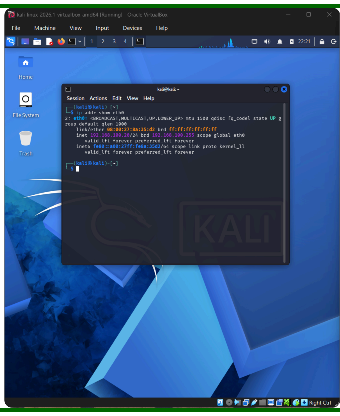
Screenshot 2

lab01_ubuntu_ip.png



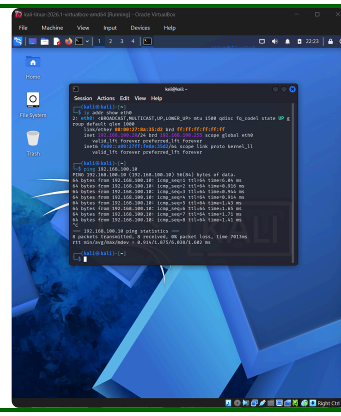
Screenshot 3

lab01_kali_ip.png



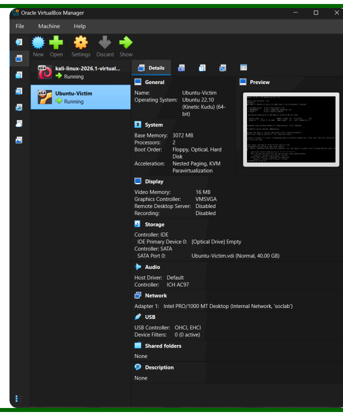
Screenshot 4

lab01_ping_success.png



Screenshot 5

lab01_vms_running.png



LAB SUMMARY

Write your own summary after completing the lab. Two things only.

What is happening in this lab — explain it in plain English:

In this first lab I started off by building my own isolated lab environment on my own laptop using VirtualBox. I set up 2 virtual machines - Kali Linux as the attacker and Ubuntu Server as the victim - and I connected them on a private network called "soclab". I manually configured a static IP address for both machines so they can find each other. I also verified that they can communicate by successfully ping test showing 0% packet loss. This is the foundation for every future lab where I start to simulate real attacks and detect and report them.

One new thing I learned that I did not know before:

I learned how to set up two virtual machines and put them on the same isolated network with static IP addresses so they can find and communicate with each other. I never knew that servers use fixed IP addresses instead of automatic ones, or why that matters for security monitoring.

COMPLETION CHECKLIST

10 / 10 complete 100%

- [x] ~~VirtualBox 7.2+ installed~~
- [x] ~~Kali Linux imported and booting successfully~~
- [x] ~~Ubuntu Server 22.04 installed with OpenSSH~~
- [x] ~~Both VMs set to Internal Network: soclab~~
- [x] ~~Ubuntu static IP 192.168.100.10 confirmed~~
- [x] ~~Kali static IP 192.168.100.20 confirmed~~
- [x] ~~Ping test — 0% packet loss confirmed~~
- [x] ~~All 5 screenshots uploaded~~
- [x] ~~Lab summary written~~
- [x] ~~Lab sheet saved and committed to GitHub~~

GITHUB COMMIT CHECKLIST

- [x] ~~screenshots/lab01_kali_desktop.png~~
- [x] ~~screenshots/lab01_ubuntu_ip.png~~
- [x] ~~screenshots/lab01_kali_ip.png~~
- [x] ~~screenshots/lab01_ping_success.png~~
- [x] ~~screenshots/lab01_vms_running.png~~
- [x] ~~lab-sheets/Lab_01_Environment_Setup.html (this file)~~
- [x] ~~configs/ubuntu_netplan.yaml~~
- [x] ~~configs/kali_interfaces.txt~~

↳ Up Next: Lab 02 — Metasploitable Setup & Network Verification

You will add Metasploitable 2 — an intentionally vulnerable VM — to your lab network, verify all three machines communicate, and run your first real Nmap scan to discover open ports. Shut down both VMs cleanly before closing — run `sudo shutdown now` in each terminal.